Annexure - A

AMITY UNIVERSITY
UTTAR PRADESH

AMITY INSTITUTE OF APPLIED SCIENCES

Synopsis of Term Paper:

Title: Statistical Analysis of Voter's Participation in Lok Sabha Elections in India

Name of Guide: Dr. Dheeraj Pawar

Programme: B.Stat (Honours/Research)		**Year/Semester:-** 3	
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Summary:-

Resource requirement:- Laptop, Access to Internet Connection, Election Commission of India website etc.

Schedule of work completion:- Commencement date: 14/05/2024

Completion date: 05/07/2024

Signature of Student

Signature of Guide

Signature of Programme Leader

Name: Dr. Niraj Kumar Singh

Approval by Board of Faculty

Member	**Signature**	**Remark (Approved / Not Approved)**

SUMMARY

Voter Participation in Lok Sabha Elections in India has been a subject of significant analysis and discussion over the years as it plays an important role in shaping the future of the nation. Voter turnout in India has shown fluctuations since the first general elections in 1951. Initially, the turnout was relatively low, but it has generally seen an upward trend over the decades. Initiatives by the Election Commission of India (ECI), such as voter education programs, improved accessibility, and voter-friendly measures, have contributed to higher voter participation. There are considerable regional differences in voter turnout, with some states consistently showing higher participation rates than others. Economic status, literacy rate and urbanization also play a major role in defining voter turnout in an area.

This report provides highly useful insights and information about voter turnout in the regions and the factors affecting it.

Annexure - B

Statistical Analysis of Voter's Participation in Lok Sabha Elections in India

TERM PAPER

Submitted to the Amity University Uttar Pradesh In partial fulfilment of requirements for the award of the Degree of

B.Stat.(Honours / Research)[2023-2027]

By

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Annexure - C

DECLARATION

I, Aksh Modgill , student of B.Stat (Honours / Research) hereby declare that the Term Paper titled "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India" which is submitted by me to Department of Statistics , Amity Institute of Applied Sciences, Amity University, Uttar Pradesh, Noida, in partial fulfillment of requirement for the award of the degree of B.Stat(Honours / Research) [2023-2027] , has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition.

Noida

Date : 10/07/2024

Aksh Modgill

Name and Signature of student

CERTIFICATE

On the basis of declaration submitted by Aksh Modgill, student of B.Stat (Honours / Research), I hereby certify that the Term Paper titled "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India" which is submitted to Department of Statistics, Amity Institute of Applied Sciences, Amity University, Uttar Pradesh, Noida, in partial fulfillment of requirement for the award of the degree of B.Stat(Honours / Research) [2023-2027], is a faithful record of work carried out by him under my guidance and supervision.

To the best of my knowledge, this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

Noida

Date : 10/07/2024

Dr. Dheeraj Pawar

(Guide)

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ACKNOWLEDGEMENT

I would like to express my gratitude to everyone who has supported and helped me to complete the NTCC report on the topic “Statistical Analysis of Voter’s Participation in Lok Sabha Elections in India“.

Firstly, I would like to thank my faculty guide Dr. Dheeraj Pawar for their guidance, advice and continuous support throughout this project. Their insightful feedback and suggestions have been helpful for the completion of this work.

I am also grateful to Amity University, Noida for giving me the opportunity to do the research work on this topic which helped me in getting extensive knowledge about this field.

Finally, I acknowledge the efforts of all those who have directly or indirectly contributed to this work. Your support is greatly appreciated.

Thank You

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ABSTRACT

This Report delves into the *Statistical Analysis of Voter's Participation in Lok Sabha Elections in India* with the aim of analyzing the trends of voter turnout in election over the years and to provide some insights about the trends and also about pattern of voting by different genders.

The methodology consisted of data collection , analysis by applying different statistical measures such as measures of central tendency ,regression analysis etc., creating graphs based on the data collected and finally writing down the conclusions and facts thus obtained from the data analysis. The data was collected from the Election Commission of India website and voter turnout app and the data analysis was done by using analysis ToolPak in Microsoft Excel.

The graphs and findings reveal a fluctuating trend in voter turnout with significant variations in some states. The voter turnout was higher in the southern and eastern part of India as compared to the northern and eastern part of India. Less than 40% of first time voters registered for the election. The major factors that affected voting percentage were socioeconomic factors, campaigns focused on creating awareness among people to vote for the country's future.

Based on the findings it is recommended to create more voter awareness and education programmes and to create some reforms to make voting easy and accessible in densely populated areas.

The insights from this research contribute to the broader understanding about voter turnout trends and patterns in Lok Sabha Elections in India.

CHAPTER 1: Introduction

India is the world's largest democracy and voter participation plays an important and crucial role in shaping the governance of the nation. *"The ignorance of one voter in a democracy impairs the security of all."- John F. Kennedy* former U.S. President, the quote very clearly explains the weightage of each voter and the importance of everyone voting in the elections.

This report aims to identify the voting trends and patterns about the voter's participation in the Lok Sabha Election and it's general impact on forming the government and giving direction to the country's future. In recent years, voter turnout in India has shown significant variation across different states. Understanding these variations is crucial for identifying the major factors that influence voter behaviour and for formulating strategies to enhance voter participation.

The report aims to provide a thorough analysis of voter participation in India by examining different aspects affecting it. The report also highlights the socioeconomic factors such as poverty etc. and the role of youth and first time voters in the elections.

At last, there are final conclusions and through the analysis, some possible solutions to the existing problems in voter participation are also specified.

CHAPTER 2 : Data Collection & Analysis

The voter turnout data for the Lok Sabha Election was collected from the website of the Election Commission of India and through the voter turnout app. The voter turnout data was collected for different states for this year's and 2019 Lok Sabha Election. The total voter turnout percentage for all the Lok Sabha elections from 1951 was also collected .

At first glance we can see that highest voter turnout was experienced in Lakshadweep with a significant voting percentage of 84.16 ,it was also the highest in the 2019 Lok Sabha election. It was followed by Assam with 81.56 voting percentage , which became the second highest voter turnout state overtaking the previous year's Nagaland. The steady growth of Assam in voter turnout can be linked to the large number of voting awareness programmes , amongst the most major programme was one in DHSK College , Dibrugarh which was done in partnership with Ministry of Youth Affairs and Sports , Government of India. The reason for Lakshadweep being the highest in voter turnout is mainly due to the small population which makes it easier to organize and encourage the voters . Being the smallest populated territory of India the community there is closely knit ,thus contributing in large number of voter turnout.

On the other hand, the lowest voter turnout was seen in the state of Bihar being merely 56.18 , which replaced Jammu & Kashmir in the previous year's Lok Sabha Election. The main reasons for this is due to the literacy rate in Bihar being the lowest and poverty being the highest in the entire nation. Despite the awareness programmes such as SVEEP (Systematic Voters' Education and Electoral Participation) etc. , disinterest amongst voters is still seen. Whereas , the same awareness programme SVEEP had a remarkable impact on the voters of Jammu & Kashmir resulting in rise of 13.61% of voter participation in 2024 elections.

The youth and the first time voter also performed poorly in the 2024 elections with just *1.8 crore new voters out of 4.9 crore having registered on the ECI website.* Telangana was the state with highest number of voters between 18-19 years. Whereas, Bihar was the the state with lowest number of first time voters with only *9.3 lakh enrolled from a potential 54 lakh voters* as reported by 'Deccan Herald' journal.

TABLE 2.1 :

Sl.No.	State/UT	No. ofPCs	Male	Female	Others	Total
1	Andaman & Nicobar Islands	1	64.41	63.77	50	64.1
2	Andhra Pradesh	25	81.04	80.3	44.34	80.66
3	Arunachal Pradesh	2	75.62	79.67	40	77.68
4	Assam	14	81.42	81.71	18.81	81.56
5	Bihar	40	53.28	59.39	6.4	56.19
6	Chandigarh	1	68.67	67.25	77.14	67.98
7	Chhattisgarh	11	73.4	72.23	29.92	72.81
8	Dadra & Nagar Haveli andDaman & Diu	2	69.99	72.73		71.31
9	Goa	2	75.42	76.66	75	76.06
10	Gujarat	25	63.52	56.56	30.77	60.13
11	Haryana	10	65.97	63.49	18.2	64.8
12	Himachal Pradesh	4	69.19	72.64	77.14	70.9
13	Jammu and Kashmir	5	60.69	56.38	31.33	58.58
14	Jharkhand	14	63.79	68.67	38.48	66.19
15	Karnataka	28	71.12	70.16	21.47	70.64
16	Kerala	20	70.63	71.88	40.87	71.27
17	Ladakh	1	71.44	72.2		71.82
18	Lakshadweep	1	82.88	85.47		84.16
19	Madhya Pradesh	29	69.37	64.24	47.17	66.87
20	Maharashtra	48	63.45	59.04	25.35	61.33
21	Manipur	2	77.63	78.72	46.77	78.19
22	Meghalaya	2	74.35	78.8	100	76.6
23	Mizoram	1	58.15	55.67		56.87
24	Nagaland	1	57.55	57.9		57.72
25	NCT OF Delhi	7	59.03	58.29	28.01	58.69
26	Odisha	21	73.37	75.55	22.92	74.44
27	Puducherry	1	78.64	79.13	69.54	78.9
28	Punjab	13	63.27	62.28	36.22	62.8
29	Rajasthan	25	62.27	60.72	53.03	61.53
30	Sikkim	1	79.93	79.84	66.67	79.88
31	Tamil Nadu	39	69.59	69.86	32.08	69.72
32	Telangana	17	66.07	65.29	30.25	65.67
33	Tripura	2	81.29	80.57	56.52	80.93
34	Uttar Pradesh	80	56.65	57.24	12.22	56.92
35	Uttarakhand	5	55.96	58.58	29.49	57.22
36	West Bengal	42	78.43	80.18	31.98	79.29
All India		542	65.8	65.78	27.08	65.79

TABLE 2.2:

1	**Lok Sabha Election 2019**	
2	**States**	**Voting Percentage**
3	Andaman & Nicobar Islands	65.12
4	Andhra Pradesh	80.38
5	Arunachal Pradesh	82.11
6	Assam	81.6
7	Bihar	57.33
8	Chandigarh	70.61
9	Chhattisgarh	71.64
10	Dadra & Nagar Haveli	79.58
11	Daman & Diu	71.85
12	Goa	75.14
13	Gujarat	64.51
14	Haryana	70.34
15	Himachal Pradesh	72.42
16	Jammu & Kashmir	44.97
17	Jharkhand	66.8
18	Karnataka	68.81
19	Kerala	77.84
20	Ladakh	
21	Lakshadweep	85.21
22	Madhya Pradesh	71.2
23	Maharashtra	61.02
24	Manipur	82.69
25	Meghalaya	71.43
26	Mizoram	63.14
27	Nagaland	83
28	NCT OF Delhi	60.6
29	Odisha	73.29
30	Puducherry	81.25
31	Punjab	65.94
32	Rajasthan	66.34
33	Sikkim	81.41
34	Tamil Nadu	72.46
35	Telangana	62.77
36	Tripura	82.4
37	Uttar Pradesh	59.21
38	Uttarakhand	61.88
39	West Bengal	81.76

CHAPTER 3: Descriptive Statistics & Measures of Central Tendency

The overall voter turnout of this year's Lok Sabha Election ranked third among all the previous elections being 65.79 % behind 2014 and 2019 Lok Sabha Elections both being 66.4% and 67.4% respectively.

The mean of the overall voter turnout was 69.17% which was also lower by 2.15 % as compared to the previous election.

The median for this year's election came out to be 70.18% which tells us that for some states half the voter turnout percentage was less than or equal to 70.18% while for others it was more than or equal to the median.

As the overall average (mean) of voter turnout is less than the median it displays that the data is negatively skewed. The kurtosis came out to be negative meaning the distribution is Platykurtic. This shows that there are less extreme values in the data set and the voting percentage is mostly closer to the mean. Thus, there is less variability due to extreme values in the voting percentage of some states.

The standard deviation of the data is 8.55 which indicates that the voting percentages are not highly spread away from the mean. The maximum voter turnout was in Lakshadweep being 84.16 % and the minimum voter turnout was in Bihar with 56.19% overall voter turnout. Hence the range of the dataset is 27.97%.

FIGURE 3.1 :

	A	B
1	*DESCRIPTIVE STATISTICS*	
2		
3	Mean	69.17806
4	Median	70.18
5	Mode	#N/A
6	Standard Deviation	8.551952
7	Sample Variance	73.13588
8	Kurtosis	-1.28272
9	Skewness	0.010237
10	Range	27.97
11	Minimum	56.19
12	Maximum	84.16
13	Sum	2490.41
14	Count	36

CHAPTER 4: Graphs, Factual Description and Analysis

In the line graph of voter turnout in Lok Sabha Elections in 2019 and 2014, some general fluctuations can be seen in voter participation in different states of both the years. The voter turnout of the year 2019 can be seen being ahead in almost all the states as compared to the 2024 Lok Sabha Election.

The most major difference in voter turnout was seen in Nagaland where the voter turnout dropped to 57.72% as compared to the impressive 83% in 2019. On the other hand, the most major increase in voter turnout in 2024 was observed in Jammu & Kashmir where the voter turnout increased to 58.58% as compared to 44.97% in 2019.

From the graph it can be observed that the states like Karnataka, Odisha, Meghalaya, Jammu & Kashmir and Telangana were able to improve their voter participation of the previous year's election, this was done by strategically organising different voter awareness programmes and campaigns in their respective states.

One of the main reasons for the decrease in voters participation in this year's election can be linked to the fact that voting took place in seven phases. Most of the people had to look for the polling date of their respective polling stations and people living in the rural areas with no internet access had little or no information about the voting date. The other prominent reason was the weather conditions, as the weather was too hot and humid, people were fatigued to stand in the voting line and register their vote.

FIGURE 4.1 :

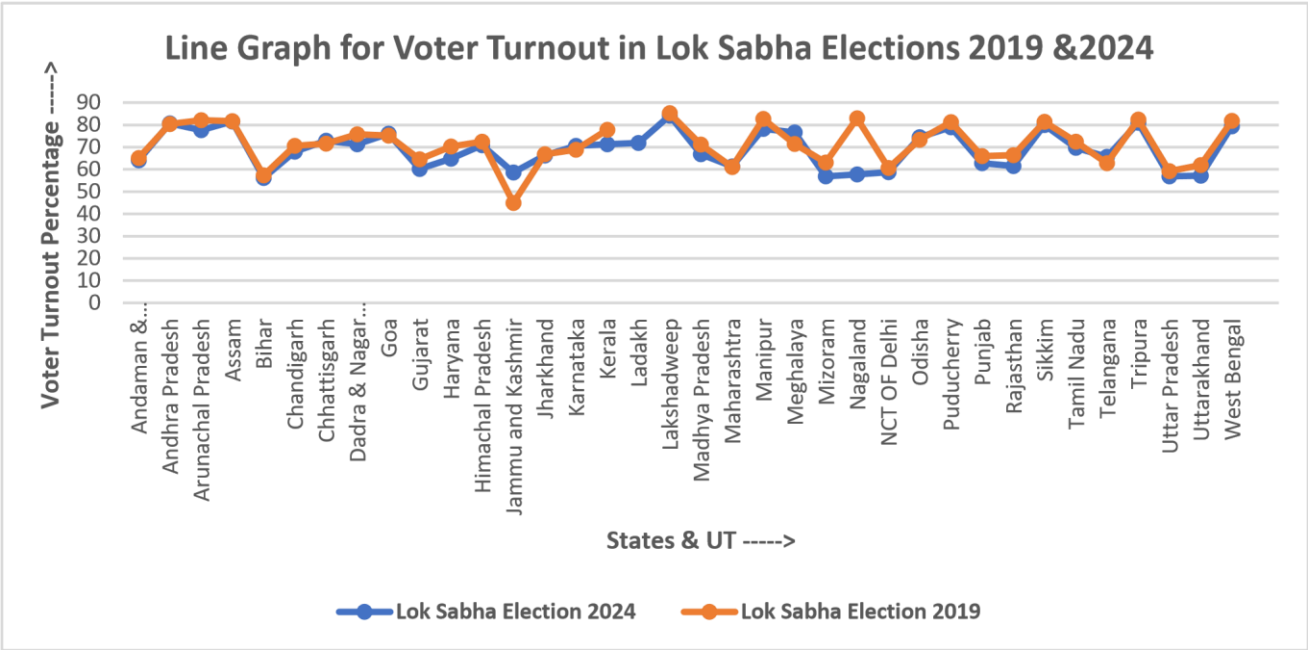
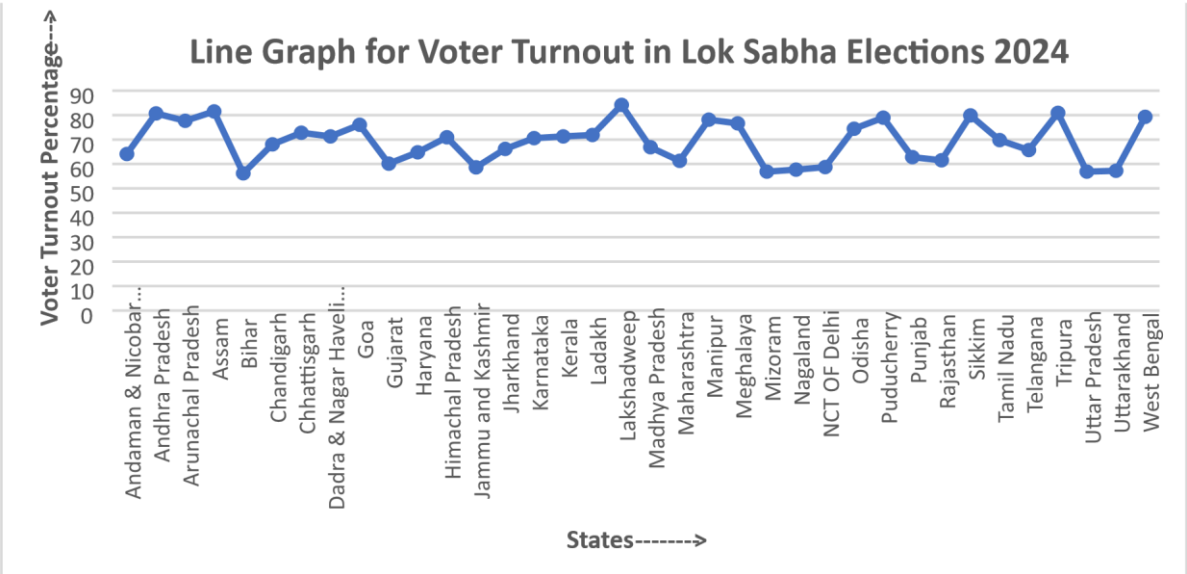


FIGURE 4.2 :

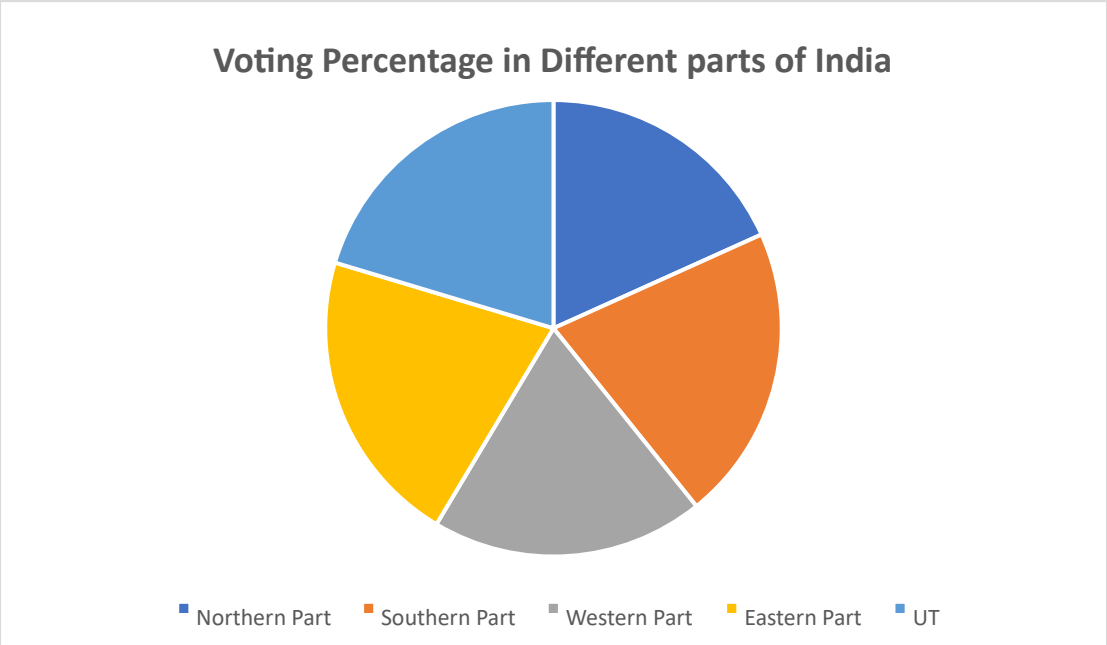


The pie chart depicts the voter's participation in different parts of India .The southern part and the eastern part can be seen having majority in the voting percentage with 71.6 and 71.98 percent respectively. Whereas, the least voter turnout can be seen in the northern region being only 62.36 % . The literacy rate was also higher in the southern part and eastern part of India .Thus , the voter turnout can also be linked to the literacy rate of that particular region. The northern part contributed to 18 % of the total chart , while western part being 19% and Union Territories being 20 % respectively. The eastern part and the southern part took the maximum area of the chart with 21% each.

TABLE 4.1 :

Voting Percentage in Different Parts of India	
Parts of India	**Voting Percentage**
Northern Part	62.36
Southern Part	71.6
Western Part	66.07
Eastern Part	71.98
UT	69.51

FIGURE 4.3 :



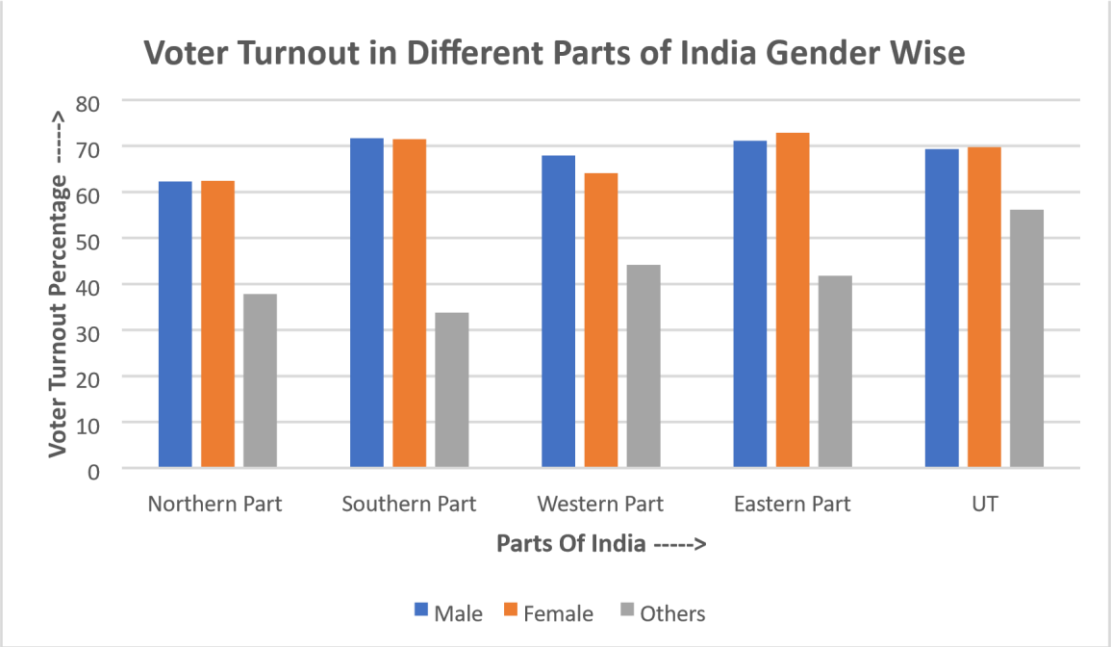
The clustered bar graph illustrates the voter turnout in different parts of India gender wise. In the bar graph, it can be seen that the voter turnout by male and female voters in most parts is approximately equal. The maximum number of female voters were seen in the eastern part of India and the maximum number of male voters were seen in the southern part of India. The maximum difference in the male and female voters was seen in the western region of India with the difference in the male and female voter turnout being 3.84%.

The increase in the number of female voters in the eastern part of India can be linked to the awareness programmes such as SVEEP (Systematic Voters' Education and Electoral Participation) etc.

TABLE 4.2: Voter Turnout in Different Parts of India Gender Wise

	Male	Female	Others
Northern Part	62.27	62.44	37.79
Southern Part	71.71	71.5	33.78
Western Part	67.92	64.08	44.12
Eastern Part	71.13	72.86	41.8
UT	69.29	69.73	56.17

FIGURE 4.4 :

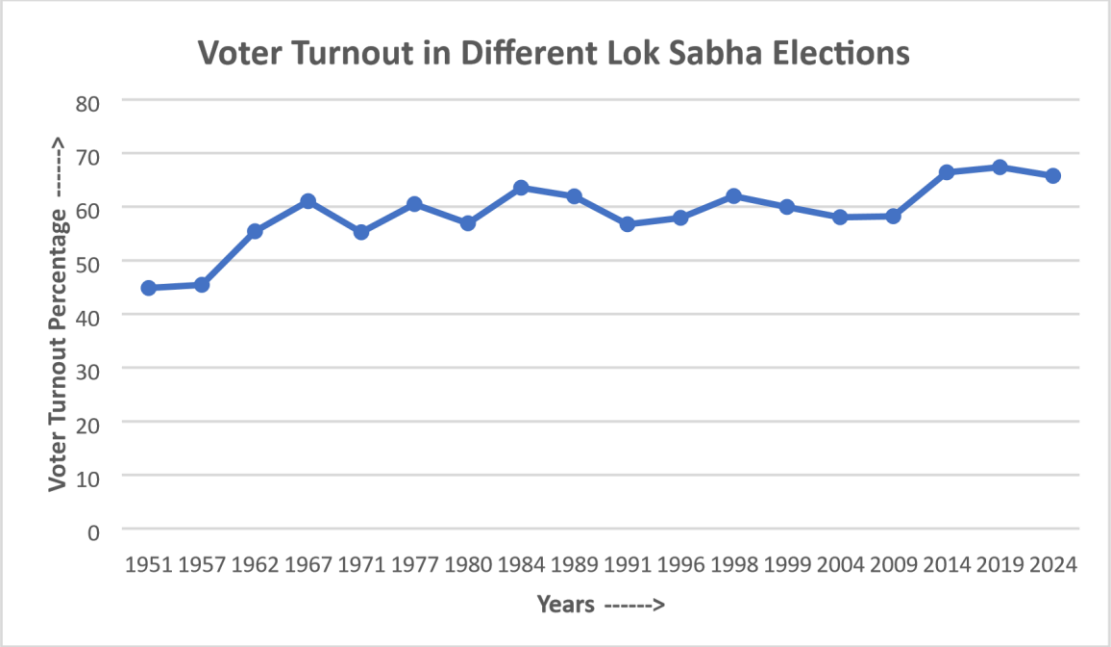


This graph shows the line graph of overall voter participation in different Lok Sabha Elections. In general, an upward trend with some fluctuations can be seen by looking at the graph . The maximum voter turnout before 2014 was in the year 1984. After that a new record was set in the year 2014 the major reason for that being Anti Incumbency Sentiment, there was a strong anti-incumbency wave against the ruling party. In 2019 the voter turnout was the highest because of the strong political campaigns and digital campaigns to influence the voters to come in large numbers to vote. In 2024 a slight dip was seen in the voting participation in comparison to the previous year.

TABLE 4.3 :

1	Election Year	Total Voter turnout %
2	1951	44.87
3	1957	45.44
4	1962	55.42
5	1967	61.04
6	1971	55.27
7	1977	60.49
8	1980	56.92
9	1984	63.56
10	1989	61.95
11	1991	56.73
12	1996	57.94
13	1998	61.97
14	1999	59.99
15	2004	58.07
16	2009	58.21
17	2014	66.4
18	2019	67.4
19	2024	65.79

FIGURE 4.5 :



CHAPTER 5: Regression Analysis

To conduct the regression analysis the dependent variable was taken to be the voter turnout percentage and the independent variable was taken as the different years of Lok Sabha Elections. The regression statistics came out to be as follows:-

- While doing the regression analysis through Microsoft Excel ToolPak we got the first value as Multiple R which is also known as Pearson's correlation coefficient value . It came out to be 0.7746 meaning a positive correlation. A positive correlation of 0.7746 means that there exists a very strong linear relationship .

- We also got R square value which is the coefficient of determination. The R square value tells us how much variance the dependent variable can be accounted for the independent variable. This value came out to be 0.6 i.e. 60% , this tells us that 60% of the voter turnout percentage can be accounted for by the years of Lok Sabha Elections. The other 40% of the variance is hence caused by other factors such as population , literacy rate etc.

- Up next, we got Standard Error which tells us about the average distance that the observed values fall from the regression line. We got this value to be 3.99%, this means that the observed voter turnout percentages were 3.99% from the regression line. The standard error being 3.99% means that there is a small standard error which indicates that the regression model and analysis is quite accurate.

- Finally we got the number of observations along with all the other measurements and it came out to be 18.

  There was also the significance F value in the analysis, which can help us in telling whether or null hypothesis is rejected or accepted. The significance F value came out to be 0.00016. By taking the null hypothesis that there is '*no linear relationship between the voter turnout and years of Lok Sabha Elections*' we can find out whether our null hypothesis is accepted or rejected. As the value of significance F is smaller than 0.05, therefore our null hypothesis is rejected. Which means that there is a linear relationship between the voter turnout and years of Lok Sabha election.

The other result we got was the coefficient table. The first row displayed the results for the intercept, which is also the point where the regression line crosses the Y-Axis when the value of the independent variable is zero. The second row displays the result for the slope . So , taking the line equation to be: y = mx + c. Here y is the predicted value of the voter turnout (dependent variable) , m is the slope of the line, x is the value of the year (independent variable) and c is the intercept. Using the information given in the table we get the equation as :

y = 0.220765.x - 380.231

By using this equation we can also predict the voter turnout in the 2029 Lok Sabha Elections :

y = 0.220765(2029) – 380.231

y=67.70%

Therefore, the expected voter turnout in the 2029 Lok Sabha Election based on previous years election is 67.70%.

TABLE 5.1 :

1	Election Year	Total Voter turnout %
2	1951	44.87
3	1957	45.44
4	1962	55.42
5	1967	61.04
6	1971	55.27
7	1977	60.49
8	1980	56.92
9	1984	63.56
10	1989	61.95
11	1991	56.73
12	1996	57.94
13	1998	61.97
14	1999	59.99
15	2004	58.07
16	2009	58.21
17	2014	66.4
18	2019	67.4
19	2024	65.79

FIGURE 5.1 :

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.774609
R Square	0.600019
Significance F	0.00016
Standard Error	3.99462
Observations	18

FIGURE 5.2 :

	Coefficients	*andard Err*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*ower 95.0%*	*pper 95.0%*
Intercept	-380.231	89.60755	-4.24329	0.00062	-570.19	-190.271	-570.19	-190.271
Election Year	0.220765	0.045062	4.899171	0.00016	0.125238	0.316291	0.125238	0.316291

FIGURE 5.3 :

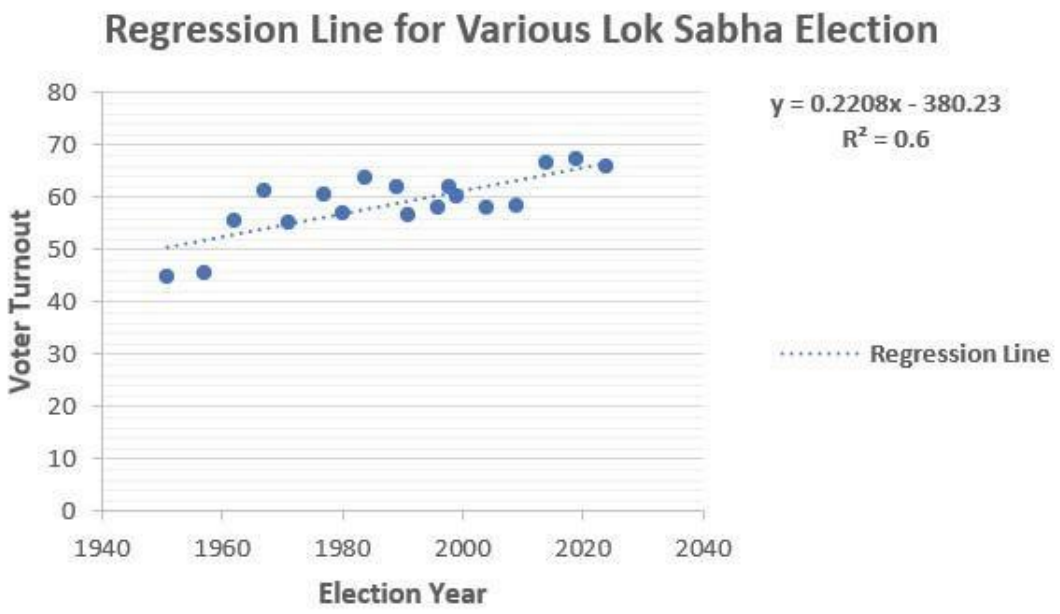
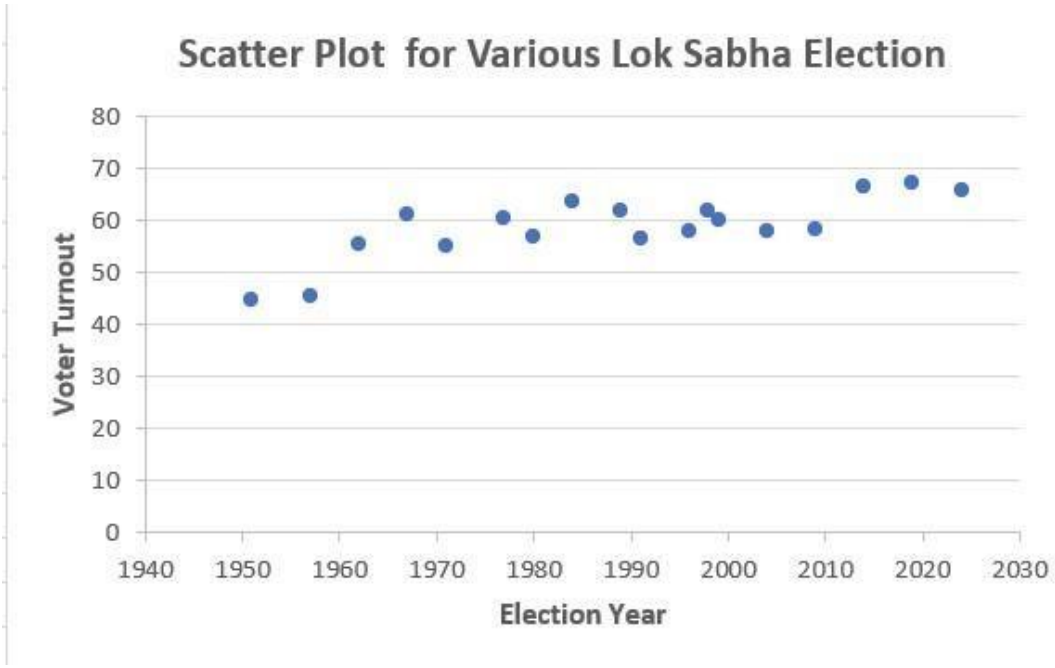


FIGURE 5.4 :



The Residual Output and table that was in the analysis tells us a predicted value of voter turnout for all the observations that were from the data entered in the regression test.

TABLE 5.2 :

RESIDUAL OUTPUT

Observation	*Total Voter*	*Residuals*
1	50.48136	-5.61136
2	51.80595	-6.36595
3	52.90978	2.510224
4	54.0136	7.0264
5	54.89666	0.373341
6	56.22125	4.268753
7	56.88354	0.036458
8	57.7666	5.793399
9	58.87042	3.079575
10	59.31195	-2.58195
11	60.41578	-2.47578
12	60.85731	1.112692
13	61.07807	-1.08807
14	62.1819	-4.1119
15	63.28572	-5.07572
16	64.38954	2.010456
17	65.49337	1.906632
18	66.59719	-0.80719

FIGURE 5.5 :

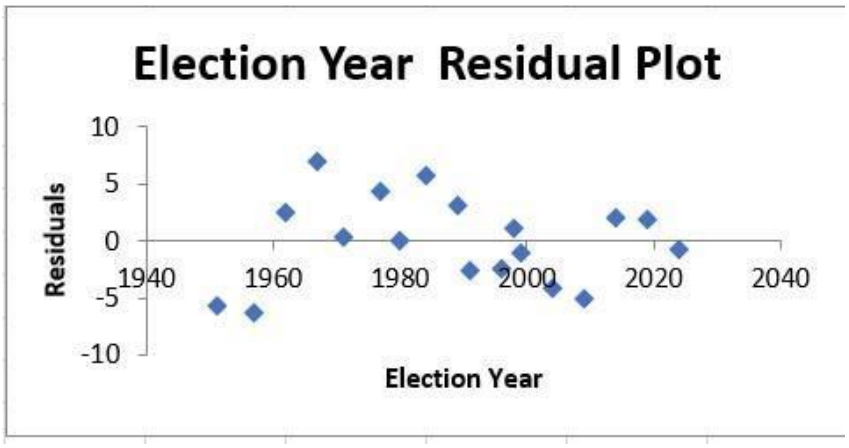
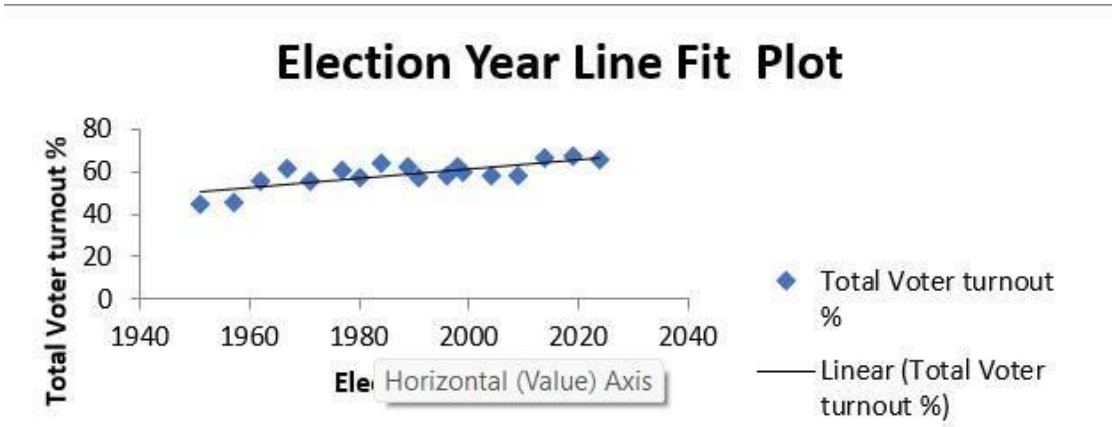


FIGURE 5.6 :

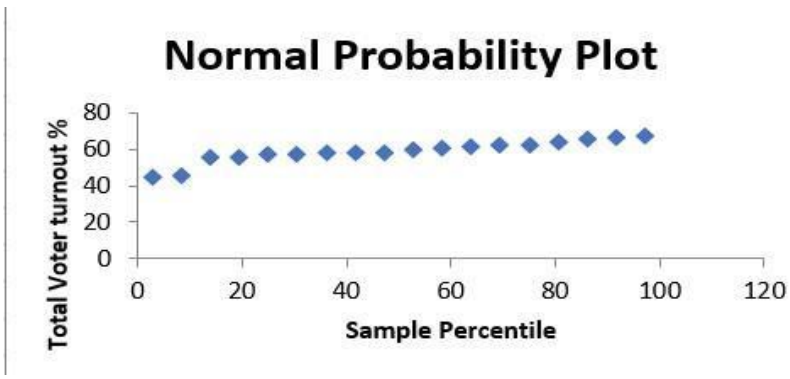


At the end, we also received a probability output table and a normal probability plot . This plot is useful to assess the normality of the dependent variable data. The straight line in the plot indicates that the voter turnout is normally distributed.

TABLE 5.3 :

PROBABILITY OUTPUT	
Percentile	*Voter turnout %*
2.777778	44.87
8.333333	45.44
13.88889	55.27
19.44444	55.42
25	56.73
30.55556	56.92
36.11111	57.94
41.66667	58.07
47.22222	58.21
52.77778	59.99
58.33333	60.49
63.88889	61.04
69.44444	61.95
75	61.97
80.55556	63.56
86.11111	65.79
91.66667	66.4
97.22222	67.4

FIGURE 5.7 :



CHAPTER 6 : Conclusions , Results & Suggestions

The overall voter turnout for Lok Sabha Election 2024 was 65.79% , which despite being lower than the election in 2019 , showcased a good amount of voter's participation in this year's election. This participation not only tells us about the citizens' commitment to shaping their future but also highlights the areas needing continued improvement to ensure even more easier and widespread voter participation. Despite the changes in vote polling patterns for example taking vote in seven phases etc., the voters seemed to be determined in doing their national work of voting.

The campaigns such as SVEEP ,campaigns organised by Election Commission of India etc. did help in the improvement of voter's participation in some states like Karnataka , Odisha , Meghalaya , Jammu & Kashmir and Telangana . While for other states like Bihar and Nagaland these campaigns seemed to have no impact at all. It's also important to note that the female participation grew in percentage in eastern part and Southern part of India as female voters were almost equal or better in these parts of the nation. The percentage of youth and first time voters remained fairly low in comparison to the previous year indicating a significant need of programmes and campaigns about voting for the younger generation. Poverty , socioeconomic factors and literacy played an important roll even in this year's election , as the majority of voters living in rural areas did not come out to vote. There have been certain positives for the states of Assam and Jammu & Kashmir as they improved their display in this year's election as compared to previous . Commuting to polling booth still remained to continue to be a problem for voters in certain cities of Kolkata , Maharashtra etc. in this year's election. It was reported by many news articles that there was heavy traffic congestion , particularly during voting days.

SUGGESTIONS :

- **Improvement in voter education and awareness:**
  The government should continue to expand and improve their voter education and awareness programmes like SVEEP , voter campaigns in schools and colleges etc. , and social media and digital platforms should be used to increase voter engagement.
- **Building awareness among first time and youth voters:** This can be done by creating awareness programmes in schools and colleges.
- **Improvement in Polling Infrastructure**:
  This can be done by investing in better polling booth infrastructure to ensure accessibility for all, particularly for persons with disabilities and the elderly.
- **Improving travel arrangements in cities :**
  A proper plan should be devised to make the travel to the polling booth hassle free and easy.

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Amity Institute Applied Sciences

WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 14/05/2024 to 20/05/2024

WPR: 1

Enrollment Number: A044161523007

Program: B. STATISTICS (HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to collect the existing datasets of voter turnouts of this year's and previous year's election to study and use it in the report.

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Collected the voter turnout data from website of Election Commission of India and Voter Turnout app.

- **FUTURE WORK PLANS: -**

  Study the data collected from online website and app.

Amity Institute Applied Sciences

WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 21/05/2024 to 27/05/2024

WPR: 2

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to study voting patterns of different states and analyze the voting percentage of different states by different genders using the data collected in the previous week.

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Collected the voter turnout data from website of Election Commission of India for phase 5 and 6 of the Lok Sabha Elections in India and studied the voter turnout percentages of different states.

- **FUTURE WORK PLANS:**
  Calculate the measures of central tendency (mean, median, mode) and standard deviation of the collected data.

Amity Institute Applied Sciences

WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 28/05/2024 to 03/06/2024

WPR: 3

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to calculate the measures of central tendency like mean, median, mode and standard deviation of this year's and previous year's voter turnout in Lok Sabha Election and analyze the voting percentage of different states .

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Collected the voter turnout data from website of Election Commission of India for phase 7 of the Lok Sabha Elections in India and calculated the mean , median , mode and standard deviation of this year's and previous year's voter turnout in Lok Sabha Election.

- **FUTURE WORK PLANS:**
  Create line graph , grouped bar chart and pie charts of voter's participation in certain regions .

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WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 04/06/2024 to 10/06/2024

WPR: 4

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to create line graph of this year's and previous year's voter turnout in Lok Sabha Election and pie chart about voter turnout in different parts of India in Lok Sabha Election.

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Created line graph of voter's participation for the Lok Sabha Election 2019 and 2024 and pie chart about voter turnout in different parts of India in Lok Sabha Election.

- **FUTURE WORK PLANS:**
  Analyze and describe the line graph and pie chart created .

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WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 11/06/2024 to 17/06/2024

WPR: 5

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to analyze and describe the line graph created of this year's and previous year's voter turnout in Lok Sabha Election and the pie chart of voter's participation in different parts of India.

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Analyzed the charts that were created and also made a brief description about the the charts and aspects of it.

- **FUTURE WORK PLANS:**
  Perform Regression Analysis with the data with the help of Microsoft Excel and find the corresponding regression equation and fit a regression line through the data points.

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WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 18/06/2024 to 24/06/2024

WPR: 6

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to perform regression analysis with the data with the help of Microsoft Excel and find the corresponding regression equation and fit a regression line through the data points .

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Collected all previous year's Lok Sabha Election total voter turnout percentage and performed regression analysis on it through Analysis ToolPak in Microsoft Excel. Calculated correlation coefficient, regression equation and created scatter plot graph with the data.

- **FUTURE WORK PLANS:**

  Predict the total voter turnout in next year's Lok Sabha Election through regression equation and write down the conclusion and facts obtained from regression analysis.

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WEEKLY PROGRESS REPORT (WPR)

For the week commencing: 25/06/2024 to 01/06/2024

WPR: 7

Enrollment Number: A044161523007

Program: B. STAT(HONOURS/RESEARCH) (2023-27)

Student Name: Aksh Modgill

Faculty Guide's Name: Dr. Dheeraj Pawar

Project Title: "Statistical Analysis of Voter's Participation in Lok Sabha Elections in India "

- **TARGETS SET FOR THE WEEK: -**
  For the commencing week, I aim to predict the total voter turnout in next year's Lok Sabha Election through regression equation and write down the conclusion and facts obtained from regression analysis.

- **PROGRESS/ACHIEVEMENTS FOR THE WEEK: -**
  Predicted the total voter turnout in next year's Lok Sabha Election through regression equation and wrote down the conclusion and facts obtained from regression analysis and started the report writing.

- **FUTURE WORK PLANS:**

  Complete and conclude the report.